1.0 Purpose

To define the Work instruction for all incidents in which standard operating procedures do not apply.

2.0 Scope

Applicable to Western Digital SDSM Penang level 2 SSD Packing Room.

3.0 Responsible Department

The process owner is responsible for managing the process and related outputs. Also the process owner is accountable for the overall process performance, ensuring the KPIs meeting targets and continual improvement.

Table 3.1 Responsibilities

Primary Owner	Approvers	Observers
Process Engineer	Process & Equipment Engineering Manager	Production Supervisor

4.0 Checklists, Reference Documents & Their Source Locations

Table 4.1: Required Checklists

Checklist No.	Source Location	Checklist Title
N/A		

Table 4.2: Reference Documents

Document No.	Source Location	Document Title
29-02-SM-02-00320	Agile	Packing VAR Matrix ESS_CSS
33-03-SM-03-00049	Agile	Turtle Diagram, Risk of Analysis, Organizational Knowledge(TRO)- SSD Equipment Engineering
33-04-SM-03-00017	Agile	Aspect Impact Assessment – SSD Equipment Engineering
33-05-SM-03-00016	Agile	HIRADC – SSD Equipment Engineering

5.0 Materials, Equipment, Tools & Safety

N/A

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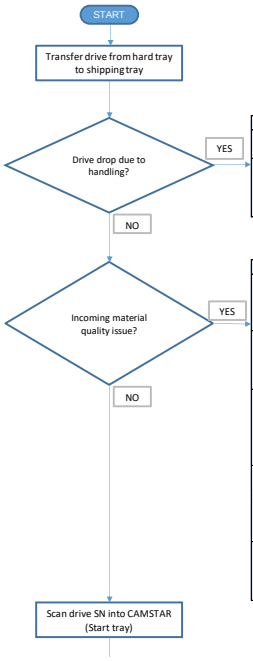
6.0 Acronyms, Terms & Abbreviations

Table 6.1: Acronyms, Terms & Abbreviations – Definitions

Acronym/Term/Abbreviation	Definition
QN	Quality Notice
OCAP	Out Of Control Action Plan
ICAP	Internal Control Action Plan
QDN	Quality Deviation Notice
QCO	Quality change Order

7.0 Requirements/Process Flow

7.1.1 Process Flow and OCAP Table of Contents for BULK pack



	Actions				
Issue	Operator	Leader	Supervisor	Technician	Engineer
Drive drop due to handling	Inform & give the dropped drive to line leader for disposition	Leader scan container defect dropped drive to PDFU	N/A	N/A	N/A
Insufficient qty	Stop line & inform line leader on insufficient drive qty issue	Get MH to top up the drive qty or change new SKUs	N/A	N/A	N/A

	Actions				
Issue	Operator	Leader	Supervisor	Technician	Engineer
Incoming packaging material issue from supplier	Stop line and inform line leader to investigate on this incoming material issue	Quarantine the affected material for disposition. Get SQE to investigate on this incoming issue.	N/A	N/A	N/A
Drive was Mixed SKUs	Stop line and inform line leader on drive Mixed SKUs	Get MH to sort out the Mixed SKUs drive and quarantine for disposition.	N/A	N/A	N/A
Drive quality issue	Stop line and inform line leader to investigate on this drive quality issue.	Quarantine the affected drive for disposition. Get QA & ME technician to investigate on this drive quality issue.	N/A	Investigate the root cause of drive quality issue.	N/A
Shipping tray was Mixed by MH	Stop line and inform line leader on shipping tray Mixed issue.	Get MH to sort out the mixed tray and quarantine for disposition.	N/A	N/A	N/A
Insufficient qty	Stop line and inform line leader on insufficient drive qty issue	Get MH to top up the drive qty or change new SKUs	N/A	N/A	N/A

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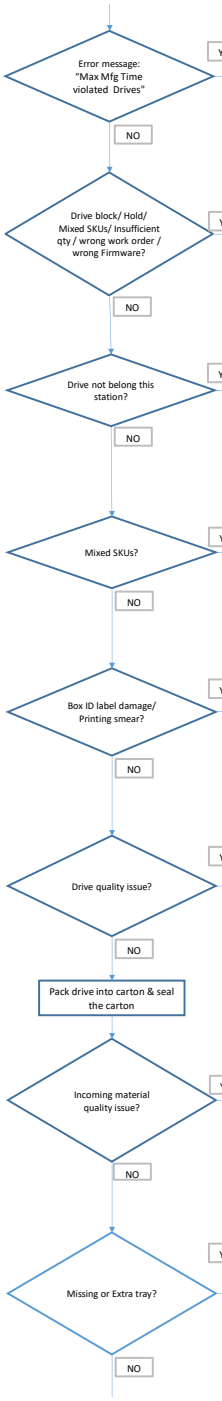

Issue	Actions: Operator	Leader	Supervisor	Technician	Engineer
Error message: "Max Mfg Time violated Drives"	Stop line and inform line leader on Max Mfg Time violated Drives issue	Reject to drives to MRB & Drives tie to PE ID to reprocess	N/A	N/A	N/A

Issue	Actions: Operator	Leader	Supervisor	Technician	Engineer
Drive was blocked/Hold/Mixed SKUs/Wrong work order/wrong firmware	Stop line & inform line leader on drive blocked / hold / Mixed SKUs / Wrong work order/wrong firmware issue	Get MH to sort out the blocked / hold / Mixed SKUs drive / Wrong work order/wrong firmware and quarantine for disposition.	Get respective owner for disposition.	N/A	N/A
Insufficient qty	Stop line & inform line leader on insufficient drive qty issue	Get MH to top up the drive qty or change new SKUs If change SKU, get MH return the partial qty to manual partial storage.	N/A	N/A	N/A

Issue	Actions: Operator	Leader	Supervisor	Technician	Engineer
Potential of duplicate SN	Stop line and inform line leader to investigate on this drive for any potential of duplicate SN	Get QA to investigate on this drive for any potential of duplicate SN	N/A	N/A	N/A

Issue	Actions: Operator	Leader	Supervisor	Technician	Engineer
Mixed SKUs into 1 tray	Stop line & inform line leader on mixed SKUs issue	Get MH to sort out the mixed SKUs drive and quarantine for disposition.	N/A	N/A	N/A

Issue	Actions: Operator	Leader	Supervisor	Technician	Engineer
Box ID label printing / damage issue	Stop line & inform line leader on Box ID label printing / damage issue.	Get OBA to dissociate the affected Box ID label. Get ME tech to check on printing / damage issue.	N/A	N/A	N/A

Issue	Actions: Operator	Leader	Supervisor	Technician	Engineer
Drive quality issue	Stop line and inform line leader to investigate on this drive quality issue.	Quarantine the affected drive for disposition. Get QA & ME technician to investigate on this drive quality issue.	N/A	Investigate the root cause of drive quality issue.	N/A

Issue	Actions: Operator	Leader	Supervisor	Technician	Engineer
Incoming packaging material issue	Stop line and inform line leader to investigate on this incoming material issue	Quarantine the affected material for disposition. Get SQE to investigate on this incoming issue.	N/A	N/A	N/A

Issue	Actions: Operator	Leader	Supervisor	Technician	Engineer
Missing / Extra shipping Tray	Stop line & inform line leader on missing or extra shipping tray issue	Stop line. Inform QA & ME technician on this quality issue. Investigate on missing / extra tray issue together with QA & ME tech. Rescreen all the carton for any missing / extra tray issue.	N/A	Investigate the root cause of missing / extra shipping tray issue.	N/A

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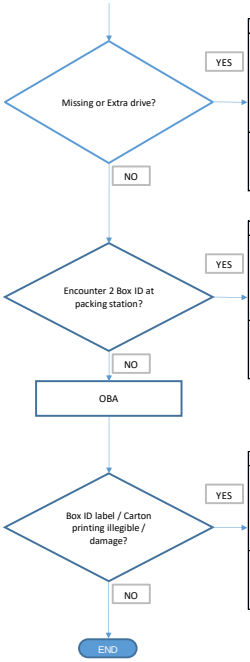
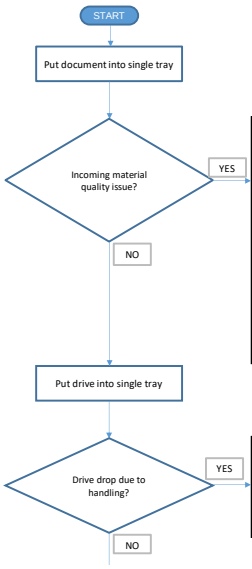


	Actions				
Issue	Operator	Leader	Supervisor	Technician	Engineer
Missing / Extra drive	Stop line & inform line leader on missing or extra drive issue	Stop line. Inform QA & ME technician on this quality issue. Investigate on missing / extra drive issue together with QA & ME tech. Rescreen all the carton for any missing / extra drive issue.	N/A	Investigate the root cause of missing / extra drive issue.	N/A

	Actions				
Issue	Operator	Leader	Supervisor	Technician	Engineer
Potential of swap / missing Box ID label	Stop line & inform line leader on 2 Box ID at packing station	Stop line. Inform QA & ME technician on this quality issue. Investigate on 2 Box ID at packing station issue together with QA & ME tech.	N/A	Investigate the root cause of potential of swap / missing Box ID label issue.	N/A

	Actions				
Issue	Operator	Leader	Supervisor	Technician	Engineer
Box ID label / Carton printing illegible / damage issue.	Stop line & inform line leader on Box ID label / Carton printing illegible / damage issue.	Stop line. Inform QA & ME technician on this quality issue. Investigate on Box ID label / Carton printing illegible / damage issue together with QA & ME tech.	N/A	Investigate the root cause of Box ID label / Carton printing illegible / damage issue.	N/A

7.1.2 Process Flow and OCAP Table of Contents for SINGLE pack



	Actions				
Issue	Operator	Leader	Supervisor	Technician	Engineer
Incoming packaging material issue supplier.	Stop line and inform line leader to investigate on this incoming material issue	Quarantine the affected material for disposition. Get SQE to investigate on this incoming issue.	N/A	N/A	N/A
Document (User manual/Warranty card) / shipping tray was Mixed	Stop line and inform line leader on document (User manual/Warranty card) / shipping tray Mixed issue.	Get MH to sort out the Mixed document (User manual/Warranty card)/ shipping tray and quarantine for disposition.	N/A	N/A	N/A
Insufficient qty	Stop line and inform line leader on insufficient document (User manual/Warranty card) / shipping tray qty issue	Get MH to top up the document (User manual/Warranty card) / shipping tray qty issue or change new SKUs if material shortage. If still insufficient qty, MH return the partial qty to material storage area.	N/A	N/A	N/A

	Actions				
Issue	Operator	Leader	Supervisor	Technician	Engineer
Drive drop due to handling	Inform & give the dropped drive to line leader for disposition	Leader scan container defect dropped drive to PDFU	N/A	N/A	N/A
Insufficient qty	Stop line & inform line leader on insufficient drive qty issue	Get MH to top up the drive qty or change new SKUs	N/A	N/A	N/A

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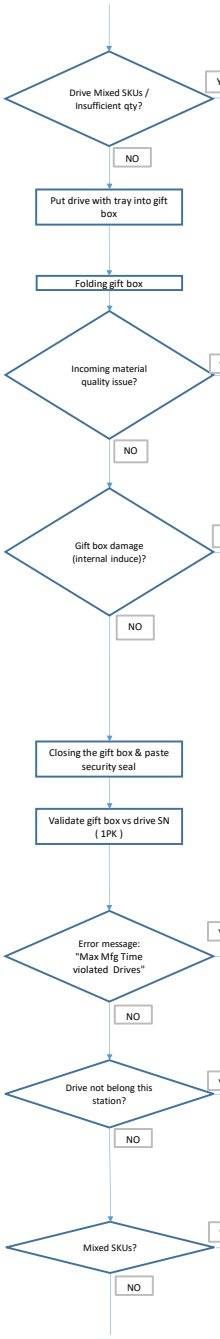

	Actions				
Issue	Operator	Leader	Supervisor	Technician	Engineer
Drive was Mixed SKUs	Stop line & inform line leader on drive Mixed SKUs issue	Get MH to sort out the Mixed SKUs drive and quarantine for disposition.	N/A	N/A	N/A
Insufficient qty	Stop line & inform line leader on insufficient drive qty issue	Get MH to top up the drive qty or change new SKUs	N/A	N/A	N/A

	Actions				
Issue	Operator	Leader	Supervisor	Technician	Engineer
Incoming packaging material issue	Stop line and inform line leader to investigate on this incoming material issue	Quarantine the affected material for disposition. Get SQE to investigate on this incoming issue.	N/A	N/A	N/A
Gift box was Mixed	Stop line and inform line leader on Gift box was Mixed issue.	Get MH to sort out the Mixed Gift box and quarantine for disposition.	N/A	N/A	N/A
Insufficient qty	Stop line and inform line leader on insufficient Gift box qty issue	Get MH to top up the gift box qty or change new SKUs if material shortage.	N/A	N/A	N/A

	Actions				
Issue	Operator	Leader	Supervisor	Technician	Engineer
Gift box damage (internal induce)	Throw away defective gift box into dustbin. Stop line & inform line leader, if gift box damage more than 5pcs	Investigate on this gift box damage issue.	N/A	N/A	N/A

	Actions				
Issue	Operator	Leader	Supervisor	Technician	Engineer
Error message: "Max Mfg Time violated Drives"	Stop line and inform line leader on Max Mfg Time violated Drives issue	Reject to drives to MRB & Drives tie to PE ID to reprocess	N/A	N/A	N/A

	Actions				
Issue	Operator	Leader	Supervisor	Technician	Engineer
Potential of duplicate SN	Stop line and inform line leader to investigate on this drive for any potential of duplicate SN	Get QA to investigate on this drive for any potential of duplicate SN	N/A	N/A	N/A

	Actions				
Issue	Operator	Leader	Supervisor	Technician	Engineer
Mixed SKUs in line	Stop line & inform line leader on mixed SKUs issue	Get MH to sort out the mixed SKUs drive and quarantine for disposition.	N/A	N/A	N/A

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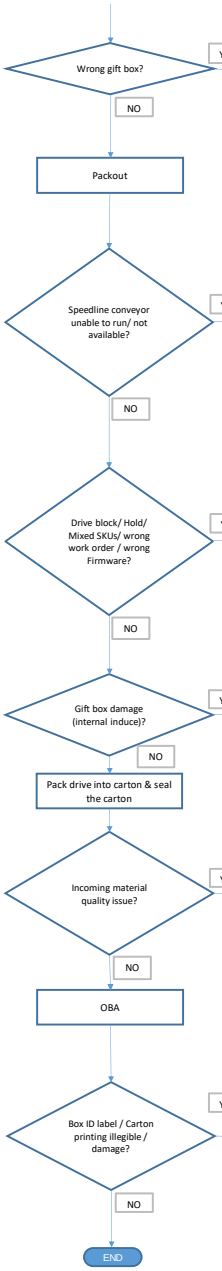

	Actions				
Issue	Operator	Leader	Supervisor	Technician	Engineer
Wrong gift box	Stop line & inform line leader on wrong gift box issue	Get MH to sort out the wrong gift box	N/A	N/A	N/A

	Actions				
Issue	Operator	Leader	Supervisor	Technician	Engineer
Speedline conveyor unable to run/ not available	Feedback to leader if speedline conveyor unable to run or not available Allow to continue running without conveyor. Follow the same headcount and process per packing VAR	Feedback conveyor unable to run issue to supervisor	Feedback conveyor unable to run issue to technician	Repair the conveyor if unable to repair internally, escalate to equipment engineer.	Get vendor help to repair the conveyor.

	Actions				
Issue	Operator	Leader	Supervisor	Technician	Engineer
Drive was blocked/Hold/Mixed SKUs/Wrong work order/wrong firmware	Stop line & inform line leader on drive blocked / hold / Mixed SKUs / Wrong work order/wrong firmware issue	Get MH to sort out the blocked / hold / Mixed SKUs drive / Wrong work order/wrong firmware. DO NOT open the gift box and quarantine at partial gift box cage for disposition.	Get respective owner for disposition.	N/A	N/A

	Actions				
Issue	Operator	Leader	Supervisor	Technician	Engineer
Gift box damage (internal induce)	Stop line & inform line leader, if gift box damage. DO NOT open the gift box.	Trigger QA to move the affected drive back to 1PK and change the gift box.	N/A	N/A	N/A

	Actions				
Issue	Operator	Leader	Supervisor	Technician	Engineer
Incoming packaging material issue from supplier	Stop line and inform line leader to investigate on this incoming material issue	Quarantine the affected material for disposition. Get SQE to investigate on this incoming issue.	N/A	N/A	N/A

	Actions				
Issue	Operator	Leader	Supervisor	Technician	Engineer
Box ID label / Carton printing illegible / damage?	Stop line & inform line leader on Box ID label / Carton printing illegible / damage issue.	Stop line. Inform QA & ME technician on this quality issue. Investigate on Box ID label / Carton printing illegible / damage issue together with QA & ME tech.	N/A	Investigate the root cause of Box ID label / Carton printing illegible / damage issue.	N/A

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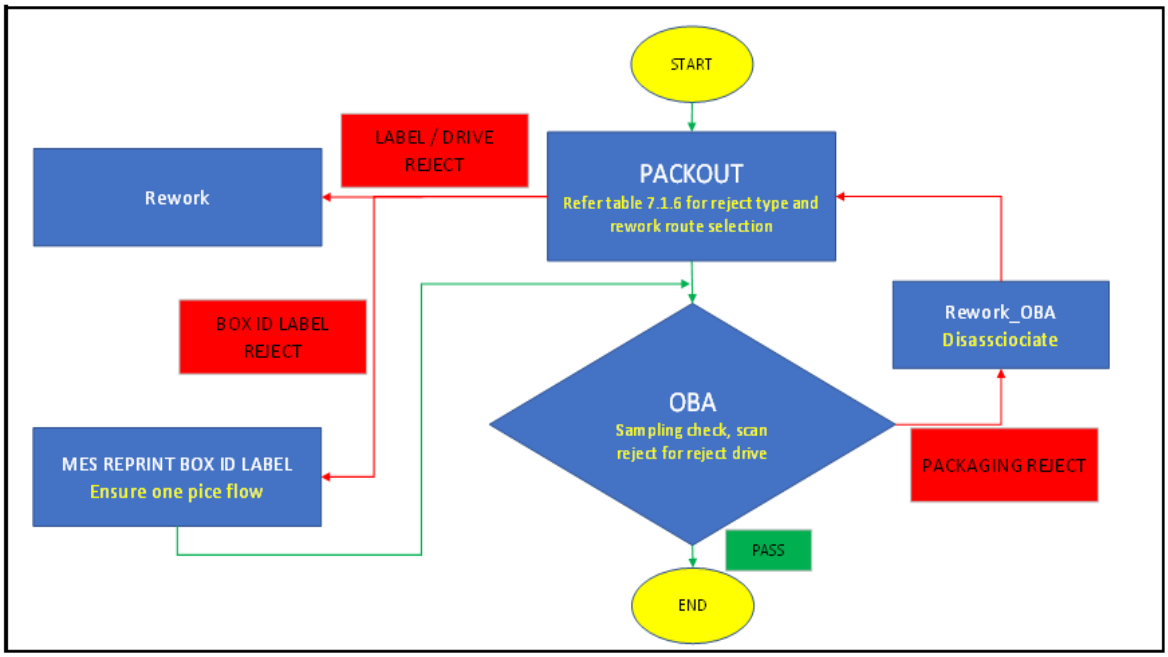
7.1.3 Inline Rework Flow for PACKOUT & OBA



7.1.4 Possible reject type and rework route

Reject type	Rework route	Rework spec code
BGA / Connector and Gold finger reject	Rework_SSMT	0903
2D label reject	Rework_Housing_2D_SN	0928
Product label reject	Rework_SMASSY2	0904
Firmware label reject	FWLABELVAL	0940
Void and marketing label reject	Rework_B_END	0907
EB reject	Rework_EBBOT	0910
Housing model OMAHA SFF reject	DEMATE	0250
Packing material reject	PACKOUT	0801

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8.0 Training Requirements

Table 8.1 Training Requirements

Training Name	Training Method	Location of Training Material	Quiz	Participants (Function/ Group)	Training Records	Monitoring Plan	Training Frequency
Out Of Control Action Plan	Instructor – led / Self study email	Agile	N/A	Technicians / Engineer	Yes	Practical Assessment	New Hire/ whenever document revise

9.0 Metrics (KPIs) & Report Distribution

Table 9.1 Metrics & Report Distribution

Metric (KPI)	Report Distribution
N/A	

10.0 Records

Table 10.1 Records

Record Name	Storage Location	Filing Index	Minimum Retention Time
Machine Downtime Report	Share drives	Daily	LI+10 Years

11.0 Key Risk Analysis

Table 11.1 Key Risk Analysis

Opportunity of Error	Process Step	Current Control
Refer to Turtle Diagram, Risk of Analysis, Organizational Knowledge(TRO)- SSD Equipment Engineering 33-03-SM-03-00049		
Refer to Aspect Impact Assessment – SSD Equipment Engineering 33-04-SM-03-00017		
Refer to HIRADC – SSD Equipment Engineering 33-05-SM-03-00016		

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12.1 Appendix

Document History – Only the last ten revision histories are to be captured here.

Rev	Originator	Description of Change
1	KE Low	Initial released of OCAP for PACKOUT & OBA
2	KE Low	Update table 7.1.4 with label damage issue , table 3.1 owner responsibility, table 11.1 key analysis , add 7.1.5 and 7.1.6
3	KK KUAN	Update table 7.1.1 & 7.1.2
4	KK KUAN	Update OCAP page 2-6

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